CULT OPEN PROJECTS 2026

Netflix Prize

Hybrid Recommender System

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315,101	**312**	**1.5M**	**98.47%**
Unique Users	Movies	Ratings	Sparsity

PROBLEM OVERVIEW

Engineering Bottlenecks in Large-Scale Recommender Systems

⚡ Extreme Sparsity

98.47% of the rating matrix is empty. Only 1.53% of interactions are observed — unstable distance computations plague naive CF models.

❄️ Cold-Start Anomaly

New users or items lack interaction history. Classical collaborative neighborhoods cannot place them without content signals.

📦 Scalability / Memory

Naive N×M matrices incur O(N·M) quadratic complexity. CSR sparse architecture reduces footprint to O(E) — linear with actual ratings.

🔍 Accuracy vs Interpretability

High-dimensional SVD factorization optimises RMSE but acts as a black box. Content models are transparent but less numerically precise.

APPROACH — THREE-MODEL PIPELINE

Raw Interaction Data → CSR Sparse Matrix → SVD Factorization → TF-IDF Content Space → Hybrid Ensemble α=0.5

Model 1 — Pure SVD

Collaborative Filtering

$$\hat{P}\_svd(u,i) = \mu + (U_u \cdot \sqrt{\Sigma}) \cdot (\sqrt{\Sigma} \cdot V^T{}_i)^T$$

- k = 10 latent factors
- Decomposes $R \approx U \times \Sigma \times V^T$
- Missing values initialised to global mean μ
- Best RMSE: 1.0406

Model 2 — Content-Based

TF-IDF + Cosine Similarity

$$\hat{P}\_cb(u,i) = \Sigma_j\ S(i,j) \cdot R_{uj}\ /\ \Sigma_j |S(i,j)|$$

- TF-IDF vectorization on titles
- Cosine similarity matrix
- Cold-start robust (fallback to mean)
- Interpretable item-level explanations

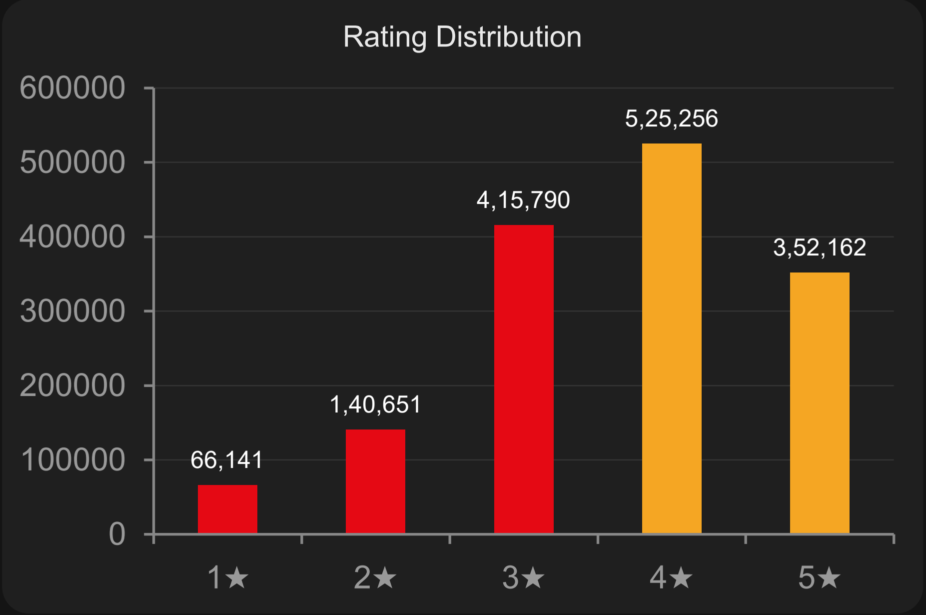
Model 3 — Hybrid Ensemble

Optimal Blending α = 0.5

$$\hat{P}\_hybrid = \alpha \cdot \hat{P}\_svd + (1-\alpha) \cdot \hat{P}\_cb$$

- Grid-search optimised α = 0.5
- Best NDCG@10: 0.0447
- Balances accuracy & cold-start
- Generates human-readable explanations

KEY INSIGHTS — EXPLORATORY DATA ANALYSIS



58.5%

Positive Ratings

4–5 star band dominates — positive platform bias inflates naive RMSE baselines.

4.8

Avg Ratings/User

Most users are sparse raters — long-tail distribution confirms need for low-rank factorization.

#1 Hit

Popularity Bias

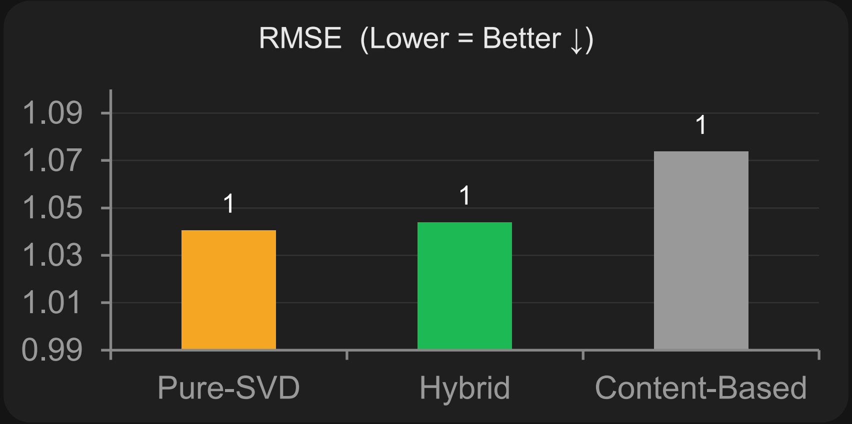
"Something's Gotta Give" alone has 118,413 ratings — blockbuster over-representation risk.

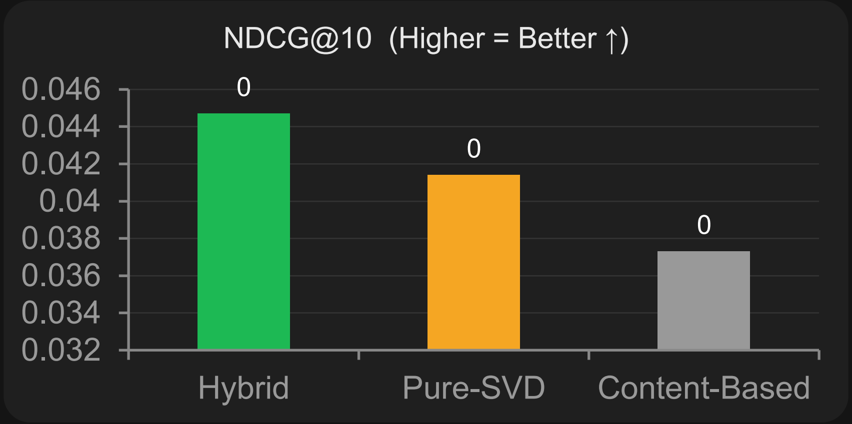
98.47%

Matrix Sparsity

Only 1.53% of the rating matrix is populated — CSR architecture is mandatory.

EXPERIMENTAL RESULTS





Model	RMSE ↓	MAE ↓	MAP@10 ↑	NDCG@10 ↑
Pure-SVD	1.0406 ★	0.8614	0.0135	0.0414
Content-Based	1.0738	0.8787	0.0167	0.0373
Hybrid ★	1.0440	0.8631	0.0156	0.0447 ★

Key Finding: Hybrid achieves best NDCG@10 — ranked recommendation quality is the primary metric for user-facing discovery.

KEY INSIGHTS

01 Extreme Sparsity Drives Degradation

98.47% of the matrix is empty. Classical neighbourhood models fail — low-rank SVD factorization is essential to maintain prediction stability across sparse validation sets.

02 Popularity Bias Dominance

Top 1% of movies account for a disproportionate share of all interactions. Latent vectors skew toward blockbusters, suppressing long-tail niche recommendations without explicit popularity penalties.

03 RMSE ≠ NDCG Optimisation

Minimising rating error does not automatically maximise ranked discovery quality. Architecture choices must align with delivery goals — Pure-SVD wins RMSE; Hybrid wins NDCG, which matters more for UI.

04 Explainability via Hybrid Layering

Pure collaborative models are black boxes. Blending a content-vector layer enables human-readable explanations ("Because you enjoyed X..."), building user trust without sacrificing accuracy.

RECOMMENDATION EXAMPLES — MODEL INTERPRETABILITY

✅ SUCCESS CASE — HIGHEST ACCURACY

User Profile ID	**135763**
Movie Asset	**Chappelle's Show: Season 1**
Ground Rating	**4** ⭐
Model Prediction	**4.00** ⭐
Prediction Error	**0.00 (Perfect Accuracy)**

🤖 AI Explanation

"Because you enjoyed 'The Game', you are likely to enjoy 'Chappelle's Show: Season 1'."

❌ FAILURE CASE — MAXIMUM DEVIATION

User Profile ID	**2296816**
Movie Asset	**Death to Smoochy**
Ground Rating	**1** ⭐
Model Prediction	**4.07** ⭐
Prediction Error	**3.07 Stars (Extreme Bias)**

🔬 Root-Cause Diagnosis

User shares high latent-vector similarity with cohort who rated positively — yet holds a sub-genre aversion undetectable from CF signals alone.

CONCLUSION & FUTURE ROADMAP

The Hybrid Ensemble (α=0.5) achieves competitive RMSE 1.0440 and best-in-class NDCG@10 0.0447 — demonstrating that blending collaborative accuracy with content interpretability delivers superior real-world recommendation quality.

NEXT-GENERATION UPGRADES

🧠 Neural CF (NCF)

GMF + MLP layers — learns non-linear interactions, reduces residual RMSE

🌐 Graph Conv. Nets

LightGCN embeddings — captures multi-hop context, mitigates sparsity

🎲 RL Bandits

UCB / Thompson Sampling — dynamic exploitation-exploration, solves cold-start

⚡ Distributed Search

Milvus / FAISS ANN indexing — sub-millisecond top-K retrieval at scale